

Learning to read and write

The ability to read and write is something that most people start to learn at a young age. As our brains develop, we learn important reading and writing skills. By the time we reach adulthood, we can read on average 200 words per minute. Reading requires several areas of the brain and body to work together. For example, when you read, your eyes need to recognize the word on a page and your brain then processes what that word says. Writing uses the brain's language areas (see pp.126–127), visual areas, and motor areas concerned with manual dexterity to make the necessary hand movements.



1 Making sounds

Babies make sounds that imitate adults but often aren't recognizable as words. This is the foundation for learning to develop language skills. Babies see and process facial expressions using the visual cortex and other areas. They then learn to associate sounds and facial expressions with things in the world.



2 Recognizing symbols

Children begin to understand what symbols mean when they are in text. They use the visual cortex and memory to translate symbols that they see into sounds. As children grow, they connect these sounds with the meanings of words and start to relate language to written text.

Reading and Writing

Our brains are hardwired for speech, but the ability to read and write is not innate. We have to start training our brains as babies to develop these complex skills.

WHAT CAUSES DYSLEXIA?

Research suggests that children with dyslexia have problems understanding the sounds letters make, but dyslexia is also found in cultures where symbols represent an idea rather than a sound.

DYSGRAPHIA

Dysgraphia is the inability to write clearly. It can be the symptom of some brain conditions, such as Parkinson's disease, that affect fine motor skills. Writing may be wobbly and indistinct or completely mangled.

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**SPEED READERS
ARE ABLE TO
READ MORE
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